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Supplementary materials to: Codimension-two bifurcation analysis of a discrete conformable fractional-order prey-predator model with nonlinear harvesting

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Supplementary Details for Section 3.1

(1) The coefficients a_{ijk}, $b_{ijk}(1 \leq i+j+k \leq 2)$in (3.2):

$$
\begin{aligned}
F_{11}(X_1, Y_1, \sigma_*) &= a_{200}X_1^2(n) + a_{110}X_1(n)Y_1(n) + a_{020}Y_1^2(n) + a_{101}X_1(n)\sigma_* \\
&\quad + a_{011}Y_1(n)\sigma_* + a_{002}\sigma_*^2 + O((|X_1(n)| + |Y_1(n)| + |\sigma_*|)^3), \\
F_{12}(X_1, Y_1, \sigma_*) &= b_{200}X_1^2(n) + b_{110}X_1(n)Y_1(n) + b_{020}Y_1^2(n) + b_{101}X_1(n)\sigma_* \\
&\quad + b_{011}Y_1(n)\sigma_* + b_{002}\sigma_*^2 + O((|X_1(n)| + |Y_1(n)| + |\sigma_*|)^3), \\
a_{100} &= 1 + \frac{x_1^* h^q}{q}\Big(-1 + \frac{\omega y_1^*}{(k+x_1^*)^2} + \frac{\overline{\sigma}}{(m+x_1^*)^2}\Big), \quad a_{010} = -\frac{h^q \omega x_1^*}{q(k+x_1^*)}, \quad a_{001} = -\frac{h^q x_1^*}{q(m+x_1^*)}, \\
a_{200} &= \frac{h^q}{q}\Big[\Big(-1 + \frac{\omega y_1^*}{(k+x_1^*)^2} + \frac{\overline{\sigma}}{(m+x_1^*)^2}\Big)\Big(1 - \frac{h^q x_1^*}{2q}\Big(-1 + \frac{\omega y_1^*}{(k+x_1^*)^2} + \frac{\overline{\sigma}}{(m+x_1^*)^2}\Big)\Big) \\
&\quad - x_1^*\Big(\frac{\omega y_1^*}{(k+x_1^*)^3} + \frac{\overline{\sigma}}{(m+x_1^*)^3}\Big)\Big], \\
a_{020} &= \frac{h^{2q}\omega^2 x_1^*}{2q^2(k+x_1^*)^2}, \quad a_{011} = \frac{\omega x_1^* h^{2q}}{q^2(m+x_1^*)(k+x_1^*)}, \quad a_{002} = \frac{x_1^* h^{2q}}{2q^2(m+x_1^*)^2}, \\
a_{110} &= \frac{h^q \omega}{q(k+x_1^*)}\Big[-\frac{h^q x_1^*}{q}\Big(-1 + \frac{\omega y_1^*}{(k+x_1^*)^2} + \frac{\overline{\sigma}}{(m+x_1^*)^2}\Big) - \frac{k}{k+x_1^*}\Big], \\
a_{101} &= -\frac{h^q}{q(m+x_1^*)} - \frac{x_1^* h^{2q}}{q^2(m+x_1^*)}\Big(-1 + \frac{\omega y_1^*}{(k+x_1^*)^2} + \frac{\overline{\sigma}}{(m+x_1^*)^2}\Big) + \frac{x_1^* h^q}{q(m+x_1^*)^2}, \\
b_{010} &= 1 - \frac{h^q \gamma\beta y_1^*}{q(k+x_1^*)}, \quad b_{001} = 0, \quad b_{100} = \frac{h^q \gamma\beta (y_1^*)^2}{q(k+x_1^*)^2}, \quad b_{002} = 0, \quad b_{101} = 0, \quad b_{011} = 0, \\
b_{200} &= \frac{h^q \beta\gamma (y_1^*)^2}{2q(k+x_1^*)^3}\Big[\frac{h^q \gamma\beta y_1^*}{q(k+x_1^*)} - 2\Big], \quad b_{110} = \frac{h^q \gamma\beta y_1^*}{q(k+x_1^*)^2}\Big[2 - \frac{h^q \gamma\beta y_1^*}{q(k+x_1^*)^2}\Big], \\
b_{020} &= \frac{h^q \gamma\beta}{2q(k+x_1^*)}\Big[\frac{h^q \gamma\beta y_1^*}{q(k+x_1^*)} - 2\Big].
\end{aligned}
$$

(2) The coefficients in (3.3) are as follows:

$$
\begin{aligned}
\widehat{F}_{12}(\widehat{X}_1(n),\widehat{Y}_1(n),\widehat{\sigma_*}) &= e_{11}\widehat{X}_1^2(n)+e_{12}\widehat{X}_1(n)\widehat{Y}_1(n)+e_{13}\widehat{Y}_1^2(n)\\
&\quad + e_{14}\widehat{X}_1\widehat{\sigma_*}+e_{15}\widehat{Y}_1\widehat{\sigma_*}+e_{16}\widehat{\sigma_*}^2+O((|\widehat{X}_1(n)|+|\widehat{Y}_1(n)|+|\widehat{\sigma_*}|)^3),\\
\widehat{F}_{12}(\widehat{X}_1(n),\widehat{Y}_1(n),\widehat{\sigma_*}) &= e_{21}\widehat{X}_1^2(n)+e_{22}\widehat{X}_1(n)\widehat{Y}_1(n)+e_{23}\widehat{Y}_1^2(n)\\
&\quad + e_{24}\widehat{X}_1\widehat{\sigma_*}+e_{25}\widehat{Y}_1\widehat{\sigma_*}+e_{26}\widehat{\sigma_*}^2+O((|\widehat{X}_1(n)|+|\widehat{Y}_1(n)|+|\widehat{\sigma_*}|)^3).\\
e_{11} &= \frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}[a_{200}a_{010}^2+a_{110}a_{010}(1-a_{100})+a_{020}(1-a_{100})^2]\\
&\quad -\frac{1}{\lambda_2-1}[b_{200}a_{010}^2+b_{110}a_{010}(1-a_{100})+b_{020}(1-a_{100})^2],\\
e_{12} &= \frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}[2a_{200}a_{010}^2+a_{110}a_{010}(\lambda_2+1-2a_{100})+2a_{020}(1-a_{100})(\lambda_2-a_{100})]\\
&\quad -\frac{1}{\lambda_2-1}[2b_{200}a_{010}^2+b_{110}a_{010}(\lambda_2+1-2a_{100})+2b_{020}(1-a_{100})(\lambda_2-a_{100})],\\
e_{13} &= \frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}[a_{200}a_{010}^2+a_{110}a_{010}(\lambda_2-a_{100})+a_{020}(\lambda_2-a_{100})^2]\\
&\quad -\frac{1}{\lambda_2-1}[b_{200}a_{010}^2+b_{110}a_{010}(\lambda_2-a_{100})+b_{020}(\lambda_2-a_{100})^2],\\
e_{14} &= \frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}[a_{101}a_{010}+a_{011}(1-a_{100})]-\frac{1}{\lambda_2-1}[b_{101}a_{010}+b_{011}(1-a_{100})],\\
e_{15} &= \frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}[a_{101}a_{010}+a_{011}(\lambda_2-a_{100})]-\frac{1}{\lambda_2-1}[b_{101}a_{010}+b_{011}(\lambda_2-a_{100})],\\
e_{16} &= \frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}a_{002}-\frac{1}{\lambda_2-1}b_{002},
\end{aligned}
$$

$$
\begin{aligned}
e_{11} &= \frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}[a_{200}a_{010}^2+a_{110}a_{010}(1-a_{100})+a_{020}(1-a_{100})^2]\\
&\quad -\frac{1}{\lambda_2-1}[b_{200}a_{010}^2+b_{110}a_{010}(1-a_{100})+b_{020}(1-a_{100})^2],\\
e_{12} &= \frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}[2a_{200}a_{010}^2+a_{110}a_{010}(\lambda_2+1-2a_{100})+2a_{020}(1-a_{100})(\lambda_2-a_{100})]\\
&\quad -\frac{1}{\lambda_2-1}[2b_{200}a_{010}^2+b_{110}a_{010}(\lambda_2+1-2a_{100})+2b_{020}(1-a_{100})(\lambda_2-a_{100})],\\
e_{13} &= \frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}[a_{200}a_{010}^2+a_{110}a_{010}(\lambda_2-a_{100})+a_{020}(\lambda_2-a_{100})^2]\\
&\quad -\frac{1}{\lambda_2-1}[b_{200}a_{010}^2+b_{110}a_{010}(\lambda_2-a_{100})+b_{020}(\lambda_2-a_{100})^2],\\
e_{14} &= \frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}[a_{101}a_{010}+a_{011}(1-a_{100})]-\frac{1}{\lambda_2-1}[b_{101}a_{010}+b_{011}(1-a_{100})],\\
e_{15} &= \frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}[a_{101}a_{010}+a_{011}(\lambda_2-a_{100})]-\frac{1}{\lambda_2-1}[b_{101}a_{010}+b_{011}(\lambda_2-a_{100})],\\
e_{16} &= \frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}a_{002}-\frac{1}{\lambda_2-1}b_{002},
\end{aligned}
$$

and $e_{2i}(1\le i\le 6)$ are obtained by replacing $\frac{\lambda_2-a_{100}}{a_{010}(\lambda_2-1)}$, $-\frac{1}{\lambda_2-1}$ in the right-hand side of $e_{1i}(1\le i\le 6)$ with $-\frac{a_{100}-1}{a_{010}(\lambda_2-1)}$ and $\frac{1}{\lambda_2-1}$, respectively.

Supplementary Details for Section 3.2

(1) The coefficients B_{00}, B_{01}, $l_i(\theta)$ $(i=1,2)$, k_{ij}, $r_{ij}(2\le i+j\le 3)$in (3.9):

$$
\begin{aligned}
B_{00} &= x_1^*\exp\Big[\frac{h^q}{q}\Big(\frac{\overline{\omega}}{\beta}-\frac{(\overline{\omega}+\omega_*)y_1^*}{k_*+\beta y_1^*}\Big)\Big]-x_1^*, \quad B_{01}=y_1^*\exp\Big[\frac{h^q\gamma k_*}{q(k_*+\beta y_1^*)}\Big]-y_1^*,\\
l_1(\theta) &= \exp\Big[\frac{h^q}{q}\Big(\frac{\overline{\omega}}{\beta}-\frac{(\overline{\omega}+\omega_*)y_1^*}{k_*+\beta y_1^*}\Big)\Big], \quad l_2(\theta)=\exp\Big[\frac{h^q\gamma k_*}{q(k_*+\beta y_1^*)}\Big], \quad k_{01}=-\frac{h^q\omega x_1^*}{q(k+x_1^*)},\\
k_{10} &= 1+\frac{x_1^*h^q}{q}\Big(-1+\frac{\omega y_1^*}{(k+x_1^*)^2}+\frac{\sigma}{(m+x_1^*)^2}\Big), \quad r_{10}=\frac{h^q\gamma\beta(y_1^*)^2}{q(k+x_1^*)^2}, \quad r_{01}=1-\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)},\\
k_{02} &= \frac{l_1(\theta)h^{2q}\omega^2x_1^*}{2q^2(k+x_1^*)^2}, \quad k_{11}=\frac{l_1(\theta)h^q\omega}{q(k+x_1^*)}\Big[-\frac{h^qx_1^*}{q}\Big(-1+\frac{\omega y_1^*}{(k+x_1^*)^2}+\frac{\sigma}{(m+x_1^*)^2}\Big)-\frac{k}{k+x_1^*}\Big],\\
k_{20} &= \frac{h^ql_1(\theta)}{q}\Big[\Big(-1+\frac{\omega y_1^*}{(k+x_1^*)^2}+\frac{\sigma}{(m+x_1^*)^2}\Big)\Big(1+\frac{h^qx_1^*}{2q}\Big(-1+\frac{\omega y_1^*}{(k+x_1^*)^2}+\frac{\sigma}{(m+x_1^*)^2}\Big)\Big)\\
&\quad -x_1^*\Big(\frac{\omega y_1^*}{(k+x_1^*)^3}+\frac{\sigma}{(m+x_1^*)^3}\Big)\Big],\\
r_{20} &= \frac{l_2(\theta)h^q\beta\gamma(y_1^*)^2}{2q(k+x_1^*)^3}\Big[\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)}-2\Big], \quad r_{11}=\frac{l_2(\theta)h^q\gamma\beta y_1^*}{q(k+x_1^*)^2}\Big[2-\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)^2}\Big],\\
r_{02} &= \frac{l_2(\theta)h^q\gamma\beta}{2q(k+x_1^*)}\Big[\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)}-2\Big], \quad k_{03}=-\frac{l_1(\theta)h^{3q}x_1^*\omega^3}{6q^3(k+x_1^*)^3},\\
k_{30} &= \frac{l_1(\theta)(h^q)^2}{6q^2}\Big(-1+\frac{\omega y_1^*}{(k+x_1^*)^2}+\frac{\sigma}{(m+x_1^*)^2}\Big)^2\Big(3+\frac{h^qx_1^*}{q}\Big(-1+\frac{\omega y_1^*}{(k+x_1^*)^2}+\frac{\sigma}{(m+x_1^*)^2}\Big)\Big)\\
&\quad -\frac{l_1(\theta)h^q}{q}\Big(\frac{\omega y_1^*}{(k+x_1^*)^3}+\frac{\sigma}{(m+x_1^*)^3}\Big)\Big(1+\frac{h^qx_1^*}{q}\Big(-1+\frac{\omega y_1^*}{(k+x_1^*)^2}+\frac{\sigma}{(m+x_1^*)^2}\Big)\Big)\\
&\quad +\frac{l_1(\theta)h^qx_1^*}{q}\Big(\frac{\omega y_1^*}{(k+x_1^*)^4}+\frac{\sigma}{(m+x_1^*)^4}\Big),\\
k_{21} &= \frac{l_1(\theta)h^q\omega}{q(k+x_1^*)}\Big[\frac{h^q}{q}\Big(-1+\frac{\omega y_1^*}{(k+x_1^*)^2}+\frac{\sigma}{(m+x_1^*)^2}\Big)\Big(\frac{x_1^*}{k+x_1^*}-1-\frac{h^qx_1^*}{q}\Big(-1+\frac{\omega y_1^*}{(k+x_1^*)^2}+\frac{\sigma}{(m+x_1^*)^2}\Big)\Big)\\
&\quad +\frac{1}{k+x_1^*}\Big(1-\frac{x_1^*}{(k+x_1^*)^2}\Big)-\frac{h^qx_1^*}{q}\Big(\frac{\omega y_1^*}{(k+x_1^*)^3}+\frac{\sigma}{(m+x_1^*)^3}\Big)\Big],\\
k_{12} &= \frac{l_1(\theta)h^{2q}\omega^2}{2q^2(k+x_1^*)^2}\Big[\frac{h^{2q}}{q^2}\Big(-1+\frac{\omega y_1^*}{(k+x_1^*)^2}+\frac{\sigma}{(m+x_1^*)^2}\Big)+\frac{k-x_1^*}{k+x_1^*}\Big],\\
r_{30} &= \frac{l_2(\theta)h^q\gamma\beta(y_1^*)^2}{6q(k+x_1^*)^4}\Big[\Big(\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)}-2\Big)\Big(\frac{h^q\gamma\beta}{q(k+x_1^*)}-3\Big)-\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)}\Big],\\
r_{21} &= \frac{l_2(\theta)h^q\gamma\beta y_1^*}{2q(k+x_1^*)^3}\Big[\Big(\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)}-2\Big)\Big(2-\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)}\Big)+\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)}\Big],\\
r_{12} &= -\frac{l_2(\theta)h^q\gamma\beta}{2q(k+x_1^*)^2}\Big[\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)}\Big(5+\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)}\Big)-2\Big],\\
r_{03} &= \frac{l_2(\theta)h^{2q}\gamma^2\beta^2}{6q^2(k+x_1^*)^2}\Big(3-\frac{h^q\gamma\beta y_1^*}{q(k+x_1^*)}\Big).
\end{aligned}
$$

(2)The coefficients $\widehat{k}_{ij}(\theta)$, $\widehat{r}_{ij}(\theta)(0\le i+j\le 2)$ in (3.10):

$$
\widehat{k}_{00}(\theta)=B_{00}, \quad \widehat{k}_{10}(\theta)=\beta r_{01}(\theta)+r_{10}(\theta)-1, \quad \widehat{k}_{01}(\theta)=-\frac{\beta q}{h^q\gamma}r_{10}(\theta)-1,
$$

$$\widehat{k}_{02}(\theta) = \frac{\beta^2 q^2}{h^{2q}\gamma^2} r_{20}(\theta), \quad \widehat{k}_{11}(\theta) = -\frac{2\beta^2 q}{h^q\gamma} r_{20}(\theta) - \frac{\beta q}{h^q\gamma} r_{11}(\theta),$$
$$\widehat{k}_{20}(\theta) = \beta^2 r_{20}(\theta) + \beta r_{11}(\theta) + r_{02}(\theta), \quad \widehat{r}_{00}(\theta) = -\frac{h^q\gamma}{\beta q} B_{00} + \frac{h^q\gamma}{q} B_{01},$$
$$\widehat{r}_{01}(\theta) = k_{10}(\theta) - \beta r_{10}(\theta) - 1, \quad \widehat{r}_{10}(\theta) = -\frac{h^q\gamma}{\beta q}\Big[\beta k_{10}(\theta) + k_{01}(\theta)\Big] + \frac{h^q\gamma}{q}\Big[\beta r_{10}(\theta) + r_{01}(\theta)\Big],$$
$$\widehat{r}_{20}(\theta) = -\frac{h^q\gamma}{\beta q}\Big[\beta^2 k_{20}(\theta) + \beta k_{11}(\theta) + k_{02}(\theta)\Big] + \frac{h^q\gamma}{q}\Big[\beta^2 r_{20}(\theta) + \beta r_{11}(\theta) + r_{02}(\theta)\Big],$$
$$\widehat{r}_{11}(\theta) = \frac{h^q\gamma}{\beta q}\Big[\frac{2\beta^2 q}{h^q\gamma} k_{20}(\theta) + \frac{\beta q}{h^q\gamma} r_{11}(\theta)\Big] - \frac{h^q\gamma}{q}\Big[\frac{2\beta^2 q}{h^q\gamma} r_{20}(\theta) + \frac{\beta q}{h^q\gamma} r_{11}(\theta)\Big],$$
$$\widehat{r}_{02}(\theta) = -\frac{\beta q}{h^q\gamma} k_{20}(\theta) + \frac{\beta^2 q}{h^q\gamma} r_{20}(\theta).$$

(3)The coefficients $a_{ij}(\theta)$, $b_{ij}(\theta)(0 \le i+j \le 2)$ in (3.11):

$$a_{00}(\theta) = \widehat{k}_{00}(\theta) - \left(\frac{1}{2}\widehat{k}_{10}(\theta) - \frac{1}{3}\widehat{r}_{01}(\theta)\right)\widehat{k}_{00}(\theta) - \left(\frac{1}{2} - \frac{1}{3}\widehat{k}_{10}(\theta) + \frac{1}{2}\widehat{r}_{01}(\theta)\right) + \frac{1}{4}\widehat{r}_{10}(\theta) - \frac{1}{3}\widehat{r}_{01}(\theta))\widehat{k}_{00}(\theta),$$
$$a_{10}(\theta) = \widehat{k}_{10}(\theta) - \frac{1}{2}\widehat{r}_{10}(\theta), \quad a_{01}(\theta) = \widehat{k}_{01}(\theta) - \frac{1}{2}\widehat{k}_{10}(\theta) + \frac{1}{3}\widehat{r}_{10}(\theta) - \frac{1}{2}\widehat{r}_{01}(\theta),$$
$$a_{20}(\theta) = \widehat{k}_{20}(\theta) - \frac{1}{2}\widehat{r}_{20}(\theta), \quad b_{11}(\theta) = \widehat{k}_{11}(\theta) - \widehat{k}_{20}(\theta) + \frac{2}{3}\widehat{r}_{20}(\theta) - \frac{1}{2}\widehat{r}_{11}(\theta),$$
$$a_{02}(\theta) = \widehat{k}_{02}(\theta) + \frac{1}{6}\widehat{k}_{20}(\theta) - \frac{1}{2}\widehat{k}_{11}(\theta) - \frac{1}{6}\widehat{r}_{20}(\theta) + \frac{1}{3}\widehat{r}_{11}(\theta) - \frac{1}{2}\widehat{r}_{02}(\theta),$$
$$b_{00}(\theta) = \widehat{r}_{00}(\theta) - \frac{1}{2}\widehat{r}_{10}(\theta)\widehat{k}_{00}(\theta) + \left(\frac{1}{3}\widehat{r}_{10}(\theta) - \frac{1}{2}\widehat{r}_{01}(\theta)\right)\widehat{r}_{00}(\theta),$$
$$b_{10}(\theta) = \widehat{r}_{10}(\theta), \quad a_{01}(\theta) = \widehat{r}_{01}(\theta) - \frac{1}{2}\widehat{r}_{10}(\theta), \quad b_{20}(\theta) = \widehat{r}_{20}(\theta),$$
$$b_{11}(\theta) = \widehat{r}_{11}(\theta) - \frac{1}{2}\widehat{r}_{20}(\theta), \quad b_{02}(\theta) = \widehat{r}_{02}(\theta) + \frac{1}{6}\widehat{r}_{20}(\theta) - \frac{1}{2}\widehat{r}_{11}(\theta).$$

(4)The coefficients $\beta_i(\theta)(i = 1,2,3,4)$in (3.14):not sure

$$\begin{aligned}\beta_1(\theta) &= \widehat{r}_{00}(\theta) + \tfrac{1}{2}\big(\tfrac{1}{3}\widehat{k}_{20}(0) - \widehat{k}_{11}(0) + 2\widehat{k}_{02}(0) - \tfrac{1}{4}\widehat{r}_{20}(0) + \tfrac{5}{12}\widehat{r}_{11}(0) \\ &\quad - \tfrac{1}{2}\widehat{r}_{02}(0)\big)\widehat{r}_{00}^2(\theta) + \tfrac{5}{12}\widehat{r}_{00}(\theta)\widehat{r}_{10}(\theta) - \tfrac{1}{2}\widehat{r}_{00}(\theta)\widehat{r}_{01}(\theta),\\ \beta_2(\theta) &= -\widehat{r}_{10}(\theta) + \widehat{r}_{01}(\theta) - (\tfrac{1}{6}\widehat{k}_{20}(0) + \tfrac{1}{2}\widehat{k}_{11}(0) - 2\widehat{k}_{02}(0) - \tfrac{1}{6}\widehat{r}_{20}(0) + \tfrac{1}{12}\widehat{r}_{11}(0))\widehat{r}_{00}(\theta),\\ \beta_3(\theta) &= \widehat{r}_{10}(\theta) - (\widehat{k}_{20}(0) - \widehat{k}_{11}(0) - \tfrac{1}{6}\widehat{r}_{20}(0))\widehat{r}_{00}(\theta),\\ \beta_4(\theta) &= \widehat{r}_{10}(\theta) + (\widehat{k}_{20}(0) - \widehat{k}_{11}(0) - \tfrac{3}{2}\widehat{r}_{20}(0) + 2\widehat{r}_{11}(0) - 2\widehat{r}_{02}(0))\widehat{r}_{00}(\theta).\end{aligned}$$

Supplementary Details for Section 3.3

(1)The coefficients $D_{ij}(\theta)$ $(i,j = 1,2)$ in (3.16) are given by:

$$\begin{aligned} D_{11}(\theta) = &\left[1 + \frac{h^q x_1^*}{q}\left(-1 + \frac{\overline{\omega} y_1^*}{(\overline{k} + x_1^*)^2} + \frac{\sigma}{(m + x_1^*)^2}\right)\right](l_1(\theta) - 1)\\ &+ \frac{h^q x_1^*}{q}\left[\frac{(\overline{\omega} + \omega_*) y_1^*}{(\overline{k} + k_* + x_1^*)^2} - \frac{\overline{\omega} y_1^*}{(\overline{k} + x_1^*)^2}\right] l_1(\theta),\end{aligned}$$

$$D_{12}(\theta)=-\frac{h^q\overline{\omega}x_1^*}{q(\overline{k}+x_1^*)}(l_1(\theta)-1)+\left[\frac{h^q\overline{\omega}x_1^*}{q(\overline{k}+x_1^*)}-\frac{h^q(\overline{\omega}+\omega_*)x_1^*}{q(\overline{k}+k_*+x_1^*)}\right]l_1(\theta),$$

$$D_{21}(\theta)=\frac{h^q\gamma\beta(y_1^*)^2}{q(\overline{k}+x_1^*)^2}(l_2(\theta)-1)+\left[\frac{h^q\gamma\beta(y_1^*)^2}{q(\overline{k}+k_*+x_1^*)^2}-\frac{h^q\gamma\beta(y_1^*)^2}{q(\overline{k}+x_1^*)^2}\right]l_2(\theta),$$

$$D_{22}(\theta)=\left[1-\frac{h^q\gamma\beta y_1^*}{q(\overline{k}+x_1^*)}\right](l_2(\theta)-1)+\frac{h^q\gamma\beta y_1^*}{q}\left[\frac{1}{\overline{k}+x_1^*}-\frac{1}{\overline{k}+k_*+x_1^*}\right]l_2(\theta).$$

Expanding $l_1(\theta)$ and $l_2(\theta)$ as Taylor series at the origin $(\omega_*,k_*)=(0,0)$ yields:

$$l_1(\theta)=1-\frac{h^q}{q\beta}\omega_*+\frac{\overline{\omega}}{\beta^2y_*}k_*+O(|\theta|^2),\quad l_2(\theta)=1+\frac{h^q\gamma}{q\beta y_*}k_*+O(|\theta|^2).$$

Obviously, $D_{11}(0)=D_{12}(0)=D_{21}(0)=D_{22}(0)=0$.

(2) The coefficients $\widetilde{k}_{ij}(\theta)$, $\widetilde{r}_{ij}(\theta)$ $(2\le i+j\le 3)$, $\widehat{D}_{ij}(\theta)(i,j=1,2)$ in (3.17):

$$\begin{aligned}
\widehat{D}_{11}(\theta)&=\frac{1}{2}[D_{11}(\alpha+1)+D_{12}\psi]-\frac{\alpha-1}{2\psi}[D_{21}(\alpha+1)+D_{22}\psi],\\
\widehat{D}_{12}(\theta)&=\frac{1}{2}[D_{11}(\alpha-1)+D_{12}\psi]-\frac{\alpha-1}{2\psi}[D_{21}(\alpha-1)+D_{22}\psi],\\
\widehat{D}_{21}(\theta)&=-\frac{1}{2}[D_{11}(\alpha+1)+D_{12}\psi]+\frac{\alpha+1}{2\psi}[D_{21}(\alpha+1)+D_{22}\psi],\\
\widehat{D}_{22}(\theta)&=-\frac{1}{2}[D_{11}(\alpha-1)+D_{12}\psi]+\frac{\alpha+1}{2\psi}[D_{21}(\alpha-1)+D_{22}\psi].\\
\widetilde{k}_{20}(\theta)&=\left(\frac{1}{2}k_{20}-\frac{\alpha-1}{2\psi}r_{20}\right)(\alpha+1)^2+\left(\frac{1}{2}k_{11}-\frac{\alpha-1}{2\psi}r_{11}\right)(\alpha+1)\psi\\
&\quad+\left(\frac{1}{2}k_{02}-\frac{\alpha-1}{2\psi}r_{02}\right)\psi^2,\\
\widetilde{k}_{11}(\theta)&=2\left(\frac{1}{2}k_{20}-\frac{\alpha-1}{2\psi}r_{20}\right)(\alpha^2-1)+2\left(\frac{1}{2}k_{11}-\frac{\alpha-1}{2\psi}r_{11}\right)\alpha\psi\\
&\quad+2\left(\frac{1}{2}k_{02}-\frac{\alpha-1}{2\psi}r_{02}\right)\psi^2,\\
\widetilde{k}_{02}(\theta)&=\left(\frac{1}{2}k_{20}-\frac{\alpha-1}{2\psi}r_{20}\right)(\alpha-1)^2+\left(\frac{1}{2}k_{11}-\frac{\alpha-1}{2\psi}r_{11}\right)(\alpha-1)\psi\\
&\quad+\left(\frac{1}{2}k_{02}-\frac{\alpha-1}{2\psi}r_{02}\right)\psi^2,\\
\widetilde{k}_{30}(\theta)&=\left(\frac{1}{2}k_{30}-\frac{\alpha-1}{2\psi}r_{30}\right)(1+\alpha)^3+\left(\frac{1}{2}k_{03}-\frac{\alpha-1}{2\psi}r_{03}\right)\psi^3\\
&\quad+\left(\frac{1}{2}k_{12}-\frac{\alpha-1}{2\psi}r_{12}\right)(1+\alpha)\psi^2+\left(\frac{1}{2}k_{21}-\frac{\alpha-1}{2\psi}r_{21}\right)(1+\alpha)^2\psi,\\
\widetilde{k}_{21}(\theta)&=3\left(\frac{1}{2}k_{30}-\frac{\alpha-1}{2\psi}r_{30}\right)(1+\alpha)^2(\alpha-1)+\left(\frac{1}{2}k_{03}-\frac{\alpha-1}{2\psi}r_{03}\right)\psi^3\\
&\quad+\left(\frac{1}{2}k_{21}-\frac{\alpha-1}{2\psi}r_{21}\right)[2(\alpha^2-1)\psi+(1+\alpha)^2\psi]\\
&\quad+\left(\frac{1}{2}k_{12}-\frac{\alpha-1}{2\psi}r_{12}\right)[2(1+\alpha)\psi^2+(\alpha-1)\psi^2],
\end{aligned}$$

$$
\begin{aligned}
\widetilde{k}_{12}(\theta) &= 3\left(\frac{1}{2}k_{30} - \frac{\alpha-1}{2\psi}r_{30}\right)(1+\alpha)(\alpha-1)^2 + \left(\frac{1}{2}k_{03} - \frac{\alpha-1}{2\psi}r_{03}\right)\psi^3 \\
&+ \left(\frac{1}{2}k_{21} - \frac{\alpha-1}{2\psi}r_{21}\right)\left[2(\alpha^2-1)\psi + (\alpha-1)^2\psi\right] \\
&+ \left(\frac{1}{2}k_{12} - \frac{\alpha-1}{2\psi}r_{12}\right)\left[(1+\alpha)\psi^2 + 2(\alpha-1)\psi^2\right], \\
\widetilde{k}_{03}(\theta) &= \left(\frac{1}{2}k_{30} - \frac{\alpha-1}{2\psi}r_{30}\right)(\alpha-1)^3 + \left(\frac{1}{2}k_{03} - \frac{\alpha-1}{2\psi}r_{03}\right)\psi^3 \\
&+ \left(\frac{1}{2}k_{21} - \frac{\alpha-1}{2\psi}r_{21}\right)(\alpha-1)^2\psi + \left(\frac{1}{2}k_{12} - \frac{\alpha-1}{2\psi}r_{12}\right)(\alpha-1)\psi^2.
\end{aligned}
$$

and $\tilde{r}_{ij}(\theta)(2 \le i+j \le 3)$ are obtained by replacing $\frac{1}{2}k_{ij}$, $-\frac{\alpha-1}{2\psi}r_{ij}$ in the right-hand side of $\tilde{k}_{ij}(\theta)(2 \le i+j \le 3)$ with $-\frac{1}{2}k_{ij}$ and $\frac{\alpha+1}{2\psi}r_{ij}$, respectively.

(3)The coefficients $\bar{k}_{ij}(\theta)$, $\bar{r}_{ij}(\theta)$ $(2 \le i+j \le 3)$, $\sigma_1(\theta)$, and $\sigma_2(\theta)$ in (3.18):

$$
\begin{aligned}
\sigma_1(\theta) &= \frac{1+\xi_2(\theta)}{2(1-\xi_1(\theta)\xi_2(\theta))}B_{00}(\theta) - \frac{(\alpha-1)+\xi_2(\theta)(\alpha+1)}{2\psi(1-\xi_1(\theta)\xi_2(\theta))}B_{01}(\theta), \\
\sigma_2(\theta) &= -\frac{1+\xi_1(\theta)}{2(1-\xi_1(\theta)\xi_2(\theta))}B_{00}(\theta) + \frac{(\alpha+1)+\xi_1(\theta)(\alpha-1)}{2\psi(1-\xi_1(\theta)\xi_2(\theta))}B_{01}(\theta). \\
\bar{k}_{20}(\theta) &= \frac{2}{1-\xi_1(\theta)\xi_2(\theta)}\left[(\tilde{k}_{20}-\xi_2\tilde{r}_{20}) + (\tilde{k}_{11}-\xi_2\tilde{r}_{11})\xi_1 + (\tilde{k}_{02}-\xi_2\tilde{r}_{02})\xi_1^2\right], \\
\bar{k}_{11}(\theta) &= \frac{1}{1-\xi_1(\theta)\xi_2(\theta)}\left[2(\tilde{k}_{20}-\xi_2\tilde{r}_{20})\xi_2 + (\tilde{k}_{11}-\xi_2\tilde{r}_{11})(1+\xi_1\xi_2) + 2(\tilde{k}_{02}-\xi_2\tilde{r}_{02})\xi_1\right], \\
\bar{k}_{02}(\theta) &= \frac{2}{1-\xi_1(\theta)\xi_2(\theta)}\left[(\tilde{k}_{20}-\xi_2\tilde{r}_{20})\xi_2^2 + (\tilde{k}_{11}-\xi_2\tilde{r}_{11})\xi_2 + (\tilde{k}_{02}-\xi_2\tilde{r}_{02})\right], \\
\bar{k}_{30}(\theta) &= \frac{6}{1-\xi_1(\theta)\xi_2(\theta)}\left[(\tilde{k}_{30}-\xi_2\tilde{r}_{30}) + (\tilde{k}_{21}-\xi_2\tilde{r}_{21})\xi_1 + (\tilde{k}_{12}-\xi_2\tilde{r}_{12})\xi_1^2 + (\tilde{k}_{03}-\xi_2\tilde{r}_{03})\xi_1^3\right], \\
\bar{k}_{21}(\theta) &= \frac{2}{1-\xi_1(\theta)\xi_2(\theta)}\Big[3(\tilde{k}_{30}-\xi_2\tilde{r}_{30})\xi_2 + (\tilde{k}_{21}-\xi_2\tilde{r}_{21})(1+2\xi_1\xi_2) \\
&+ (\tilde{k}_{12}-\xi_2\tilde{r}_{12})(2\xi_1^2\xi_2+2\xi_1) + 3(\tilde{k}_{03}-\xi_2\tilde{r}_{03})\xi_1\Big], \\
\bar{k}_{12}(\theta) &= \frac{2}{1-\xi_1(\theta)\xi_2(\theta)}\Big[3(\tilde{k}_{30}-\xi_2\tilde{r}_{30})\xi_2^2 + (\tilde{k}_{21}-\xi_2\tilde{r}_{21})(\xi_1\xi_2^2+2\xi_2) \\
&+ (\tilde{k}_{12}-\xi_2\tilde{r}_{12})(1+2\xi_1\xi_2) + 3(\tilde{k}_{03}-\xi_2\tilde{r}_{03})\xi_1\Big], \\
\bar{k}_{03}(\theta) &= \frac{6}{1-\xi_1(\theta)\xi_2(\theta)}\left[(\tilde{k}_{30}-\xi_2\tilde{r}_{30})\xi_2^3 + (\tilde{k}_{21}-\xi_2\tilde{r}_{21})\xi_2^2 + (\tilde{k}_{12}-\xi_2\tilde{r}_{12})\xi_2 + (\tilde{k}_{03}-\xi_2\tilde{r}_{03})\right].
\end{aligned}
$$

and $\bar{r}_{ij}(\theta)(2 \le i+j \le 3)$ are obtained by replacing $(\tilde{k}_{ij}-\xi_2\tilde{r}_{ij})$ in $\bar{k}_{ij}(\theta)(2 \le i+j \le 3)$ with $(-\xi_1\tilde{k}_{ij}+\tilde{r}_{ij})$.

(4)The coefficients in (3.19) are given by:

$$
\begin{aligned}
A_1 &= -\frac{h^q x_1^*}{2q\beta}, \quad A_2 = \frac{h^q x_1^* \varpi}{2q\beta^2 y_1^*} - \frac{h^q\gamma(\alpha-1)}{2\psi\beta}, \quad B_1 = \frac{h^q x_1^*}{2q\beta}, \quad B_2 = -\frac{h^q x_1^*}{2q\beta(y_1^*)^2} + \frac{h^q\gamma(\alpha+1)}{2\psi\beta}, \\
A_3 &= \frac{1}{2}\left[\frac{(\alpha+1)h^q}{q}\left(\frac{x_1^*}{y_1^*\beta} - \frac{1}{\beta}\left(1+\frac{h^q x_1^*}{q}\left(-1+\frac{\varpi}{\beta y_1^*}+\frac{\sigma}{(m+x_1^*)^2}\right)\right)\right) + \frac{\psi h^q x_1^*}{q\beta y_1^*}\left(\frac{h^q\varpi}{q\beta}-1\right)\right],
\end{aligned}
$$

$$
\begin{aligned}
A_4 &= \frac{1}{2}\left[\frac{(\alpha+1)\overline{\omega}}{\beta^2 y_1^*}\left(1+\frac{h^q x_1^*}{q}\left(-1+\frac{\overline{\omega}}{\beta y_1^*}+\frac{\sigma}{(m+x_1^*)^2}\right)-\frac{2h^q x_1^* \overline{\omega}}{q\beta^2 y_1^*}\right)+\frac{\psi h^q x_1^* \overline{\omega}}{q\beta y_1^*}\left(\frac{1}{\beta y_1^*}-\frac{\overline{\omega}}{y_1^*\beta^2}\right)\right] \\
&\quad -\frac{\alpha-1}{2\psi}\left[\frac{(\alpha-1)h^q\gamma}{q\beta}\left(\frac{h^q\gamma}{q\beta}-\frac{2}{\beta y_1^*}\right)+\frac{\psi h^q\gamma}{q}\left(\frac{1}{\beta y_1^*}\left(1-\frac{h^q\gamma}{q}\right)+\frac{1}{\beta y_1^*}\right)\right], \\
B_3 &= -\frac{1}{2}\left[\frac{(\alpha-1)h^q}{q}\left(\frac{x_1^*}{y_1^*\beta}-\frac{1}{\beta}\left(1+\frac{h^q x_1^*}{q}\left(-1+\frac{\overline{\omega}}{\beta y_1^*}+\frac{\sigma}{(m+x_1^*)^2}\right)\right)\right)+\frac{\psi h^q x_1^*}{2q\beta y_1^*}\left(\frac{h^q\overline{\omega}}{q\beta}-1\right)\right], \\
B_4 &= -\frac{1}{2}\left[\frac{(\alpha-1)\overline{\omega}}{\beta^2 y_1^*}\left(1+\frac{h^q x_1^*}{q}\left(-1+\frac{\overline{\omega}}{\beta y_1^*}+\frac{\sigma}{(m+x_1^*)^2}\right)-\frac{2h^q x_1^* \overline{\omega}}{q\beta^2 y_1^*}\right)+\frac{\psi h^q x_1^* \overline{\omega}}{q\beta y_1^*}\left(\frac{1}{\beta y_1^*}-\frac{\overline{\omega}}{y_1^*\beta^2}\right)\right] \\
&\quad +\frac{\alpha+1}{2\psi}\left[\frac{(\alpha-1)h^q\gamma}{q\beta}\left(\frac{h^q\gamma}{q\beta y_1^*}-\frac{2}{\beta y_1^*}\right)+\frac{\psi h^q\gamma}{q}\left(\frac{1}{\beta y_1^*}\left(1-\frac{h^q\gamma}{q}\right)+\frac{1}{\beta y_1^*}\right)\right].
\end{aligned}
$$

Supplementary Details for Section 3.4

(1)The coefficients $\widehat{\sigma}_{ij}(\theta)$,$\widehat{\zeta}_{ij}(\theta)$ $(2 \leq i+j \leq 3)$, and $\widetilde{D}_{ij}(\theta)(i,j=1,2)$ in (3.24):

$$
\begin{aligned}
&\widetilde{D}_{11}(\theta) = \hat{\xi}_1 D_{21} + D_{22}, \quad \widetilde{D}_{12}(\theta) = \hat{\xi}_2 D_{21}, \\
&\widetilde{D}_{21}(\theta) = \hat{\xi}_3(\hat{\xi}_1 D_{11} + D_{12}) + \hat{\xi}_4(\hat{\xi}_1 D_{21} + D_{22}), \quad \widetilde{D}_{22}(\theta) = \hat{\xi}_2(\hat{\xi}_3 D_{11} + \hat{\xi}_4 D_{21}). \\
&\widehat{\sigma}_{20}(\theta) = r_{20}\hat{\xi}_1^2 + r_{11}\hat{\xi}_1 + r_{02}, \widehat{\sigma}_{11}(\theta) = 2r_{20}\hat{\xi}_1\hat{\xi}_2 + r_{11}\hat{\xi}_2, \\
&\widehat{\sigma}_{02}(\theta) = r_{20}\hat{\xi}_2, \widehat{\sigma}_{30}(\theta) = r_{30}\hat{\xi}_1^3 + r_{21}\hat{\xi}_1^2 + r_{12}\hat{\xi}_1 + r_{03}, \\
&\widehat{\sigma}_{21}(\theta) = 3r_{30}\hat{\xi}_1^2\hat{\xi}_2 + 2r_{21}\hat{\xi}_1\hat{\xi}_2 + r_{12}\hat{\xi}_2, \widehat{\sigma}_{12}(\theta) = 3r_{30}\hat{\xi}_1\hat{\xi}_2^2 + r_{21}\hat{\xi}_2^2, \\
&\widehat{\sigma}_{03}(\theta) = r_{30}\hat{\xi}_2^3, \widehat{\zeta}_{20}(\theta) = \hat{\xi}_3(k_{20}\hat{\xi}_1^2 + k_{11}\hat{\xi}_1 + k_{02}) + \hat{\xi}_4(r_{20}\hat{\xi}_1^2 + r_{11}\hat{\xi}_1 + r_{02}), \\
&\widehat{\zeta}_{11}(\theta) = \hat{\xi}_3(2k_{20}\hat{\xi}_1\hat{\xi}_2 + k_{11}\hat{\xi}_2) + \hat{\xi}_4(2r_{20}\hat{\xi}_1\hat{\xi}_2 + r_{11}\hat{\xi}_2), \widehat{\zeta}_{02}(\theta) = \hat{\xi}_2(k_{20}\hat{\xi}_3 + r_{20}\hat{\xi}_4), \\
&\widehat{\zeta}_{30}(\theta) = \hat{\xi}_3(k_{30}\hat{\xi}_1^3 + k_{21}\hat{\xi}_1^2 + k_{12} + k_{03}) + \hat{\xi}_4(r_{30}\hat{\xi}_1^3 + r_{21}\hat{\xi}_1^2 + r_{12} + r_{03}), \\
&\widehat{\zeta}_{21}(\theta) = \hat{\xi}_3(3k_{30}\hat{\xi}_1^2 + 2k_{21}\widehat{\xi}_1\hat{\xi}_2) + \hat{\xi}_4(3r_{30}\hat{\xi}_1^2\hat{\xi}_2 + 2r_{21}\widehat{\xi}_1\hat{\xi}_2), \\
&\widehat{\zeta}_{12}(\theta) = \hat{\xi}_3(3k_{30}\hat{\xi}_1\hat{\xi}_2^2 + k_{21}\hat{\xi}_2^2) + \hat{\xi}_4(3r_{30}\hat{\xi}_1\hat{\xi}_2^2 + r_{21}\hat{\xi}_2^2), \\
&\widehat{\zeta}_{03}(\theta) = \hat{\xi}_2^3(k_{30}\hat{\xi}_3 + r_{30}\hat{\xi}_4).
\end{aligned}
$$

(2)The coefficients θ_1, θ_2, $\tilde{\sigma}_{ij}(\theta)$, and $\tilde{\zeta}_{ij}(\theta)$in (3.25):

$$
\begin{aligned}
&\theta_1 = \widetilde{D}_{21}(1+\widetilde{D}_{12}) - \widetilde{D}_{11}\widetilde{D}_{22}, \theta_2 = \widetilde{D}_{11} + \widetilde{D}_{22} - 1, \\
&\tilde{\sigma}_{20} = -\frac{\widetilde{D}_{11}}{1+\widetilde{D}_{12}}\widehat{\sigma}_{11} + \widehat{\sigma}_{20} + \frac{\widetilde{D}_{11}^2}{(1+\widetilde{D}_{12})^2}\widehat{\sigma}_{02}, \quad \tilde{\sigma}_{11} = \frac{1}{1+\widetilde{D}_{12}}\widehat{\sigma}_{11} - \frac{2\widetilde{D}_{11}}{(1+\widetilde{D}_{12})^2}\widehat{\sigma}_{02}, \\
&\tilde{\sigma}_{02} = \frac{1}{(1+\widetilde{D}_{12})^2}\widehat{\sigma}_{02}, \quad \tilde{\sigma}_{30} = \widehat{\sigma}_{30} - \frac{\widetilde{D}_{11}}{1+\widetilde{D}_{12}}\widehat{\sigma}_{21} + \frac{\widetilde{D}_{12}^2}{(1+\widetilde{D}_{12})^2}\widehat{\sigma}_{12} - \frac{\widetilde{D}_{11}^3}{(1+\widetilde{D}_{12})^3}\widehat{\sigma}_{03}, \\
&\tilde{\sigma}_{03} = \frac{1}{(1+\widetilde{D}_{12})^3}\widehat{\sigma}_{03}, \quad \tilde{\sigma}_{21} = \frac{1}{1+\widetilde{D}_{12}}\widehat{\sigma}_{21} - \frac{2\widetilde{D}_{11}}{(1+\widetilde{D}_{12})^2}\widehat{\sigma}_{12} + \frac{3\widetilde{D}_{11}^2}{(1+\widetilde{D}_{12})^3}\widehat{\sigma}_{03}, \\
&\tilde{\sigma}_{12} = \frac{1}{(1+\widetilde{D}_{12})^2}\widehat{\sigma}_{12} - \frac{3\widetilde{D}_{11}}{(1+\widetilde{D}_{12})^3}\widehat{\sigma}_{03}, \\
&\tilde{\zeta}_{20} = -\frac{\widetilde{D}_{11}}{1+\widetilde{D}_{12}}\left[\widetilde{D}_{11}\widehat{\sigma}_{11} + (1+\widetilde{D}_{12})\widehat{\zeta}_{11}\right] + \left[\widetilde{D}_{11}\widehat{\sigma}_{20} + (1+\widetilde{D}_{12})\widehat{\zeta}_{20}\right] \\
&\qquad + \frac{\widetilde{D}_{11}^2}{(1+\widetilde{D}_{12})^2}\left(\widetilde{D}_{11}\widehat{\sigma}_{02} + (1+\widetilde{D}_{12})\widehat{\zeta}_{02}\right),
\end{aligned}
$$

$$
\begin{aligned}
\tilde{\zeta}_{11} &= \frac{1}{1+\widetilde{D}_{12}}\left[\widetilde{D}_{11}\widehat{\sigma}_{11} + (1+\widetilde{D}_{12})\widehat{\zeta}_{11}\right] - \frac{2\widetilde{D}_{11}}{(1+\widetilde{D}_{12})^2}\left(\widetilde{D}_{11}\widehat{\sigma}_{02} + (1+\widetilde{D}_{12})\widehat{\zeta}_{02}\right), \\
\tilde{\zeta}_{02} &= \frac{1}{(1+\widetilde{D}_{12})^2}\left(\widetilde{D}_{11}\widehat{\sigma}_{02} + (1+\widetilde{D}_{12})\widehat{\zeta}_{02}\right), \quad \tilde{\zeta}_{03} = \frac{1}{(1+\widetilde{D}_{12})^3}\left[\widetilde{D}_{11}\widehat{\sigma}_{30} + (1+\widetilde{D}_{12})\widehat{\zeta}_{30}\right], \\
\tilde{\zeta}_{30} &= \left[\widetilde{D}_{11}\widehat{\sigma}_{30} + (1+\widetilde{D}_{12})\widehat{\zeta}_{30}\right] - \frac{\widetilde{D}_{11}}{1+\widetilde{D}_{12}}\left[\widetilde{D}_{11}\widehat{\sigma}_{21} + (1+\widetilde{D}_{12})\widehat{\zeta}_{21}\right] \\
&\quad + \frac{\widetilde{D}_{11}^2}{(1+\widetilde{D}_{12})^2}\left[\widetilde{D}_{11}\widehat{\sigma}_{12} + (1+\widetilde{D}_{12})\widehat{\zeta}_{12}\right] - \frac{\widetilde{D}_{11}^3}{(1+\widetilde{D}_{12})^3}\left[\widetilde{D}_{11}\widehat{\sigma}_{03} + (1+\widetilde{D}_{12})\widehat{\zeta}_{03}\right], \\
\tilde{\zeta}_{21} &= \frac{1}{1+\widetilde{D}_{12}}\left[\widetilde{D}_{11}\widehat{\sigma}_{21} + (1+\widetilde{D}_{12})\widehat{\zeta}_{21}\right] - \frac{2\widetilde{D}_{11}}{(1+\widetilde{D}_{12})^2}\left[\widetilde{D}_{11}\widehat{\sigma}_{12} + (1+\widetilde{D}_{12})\widehat{\zeta}_{12}\right] \\
&\quad + \frac{3\widetilde{D}_{11}^2}{(1+\widetilde{D}_{12})^3}\left[\widetilde{D}_{11}\widehat{\sigma}_{30} + (1+\widetilde{D}_{12})\widehat{\zeta}_{30}\right], \\
\tilde{\zeta}_{12} &= \frac{1}{(1+\widetilde{D}_{12})^2}\left[\widetilde{D}_{11}\widehat{\sigma}_{12} + (1+\widetilde{D}_{12})\widehat{\zeta}_{21}\right] - \frac{3\widetilde{D}_{11}}{(1+\widetilde{D}_{12})^3}\left[\widetilde{D}_{11}\widehat{\sigma}_{30} + (1+\widetilde{D}_{12})\widehat{\zeta}_{30}\right].
\end{aligned}
$$